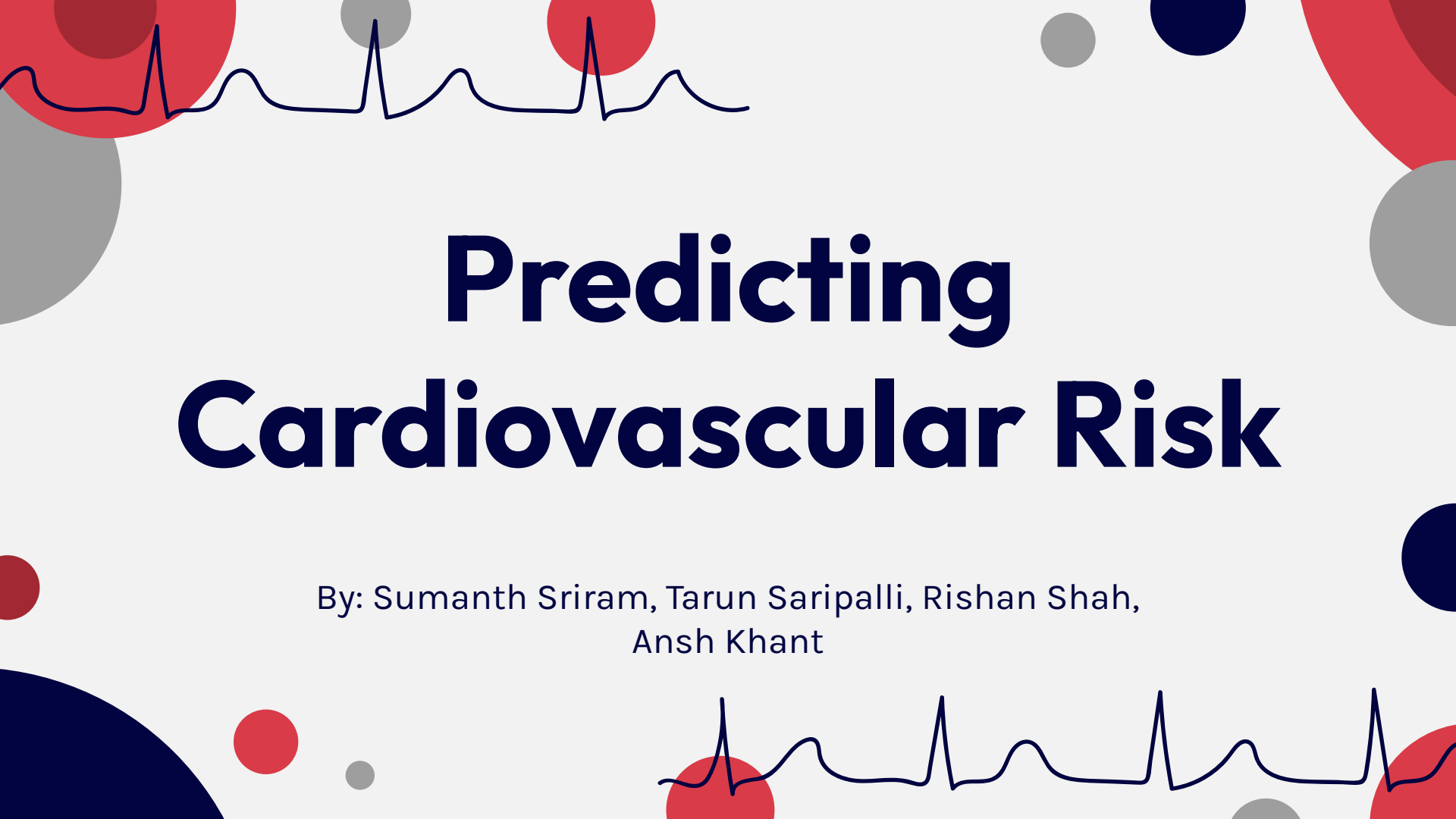


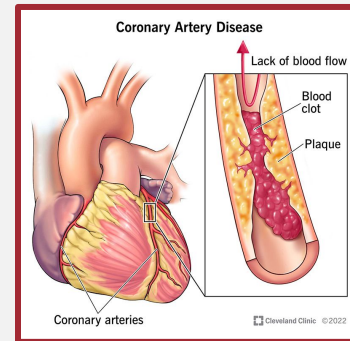


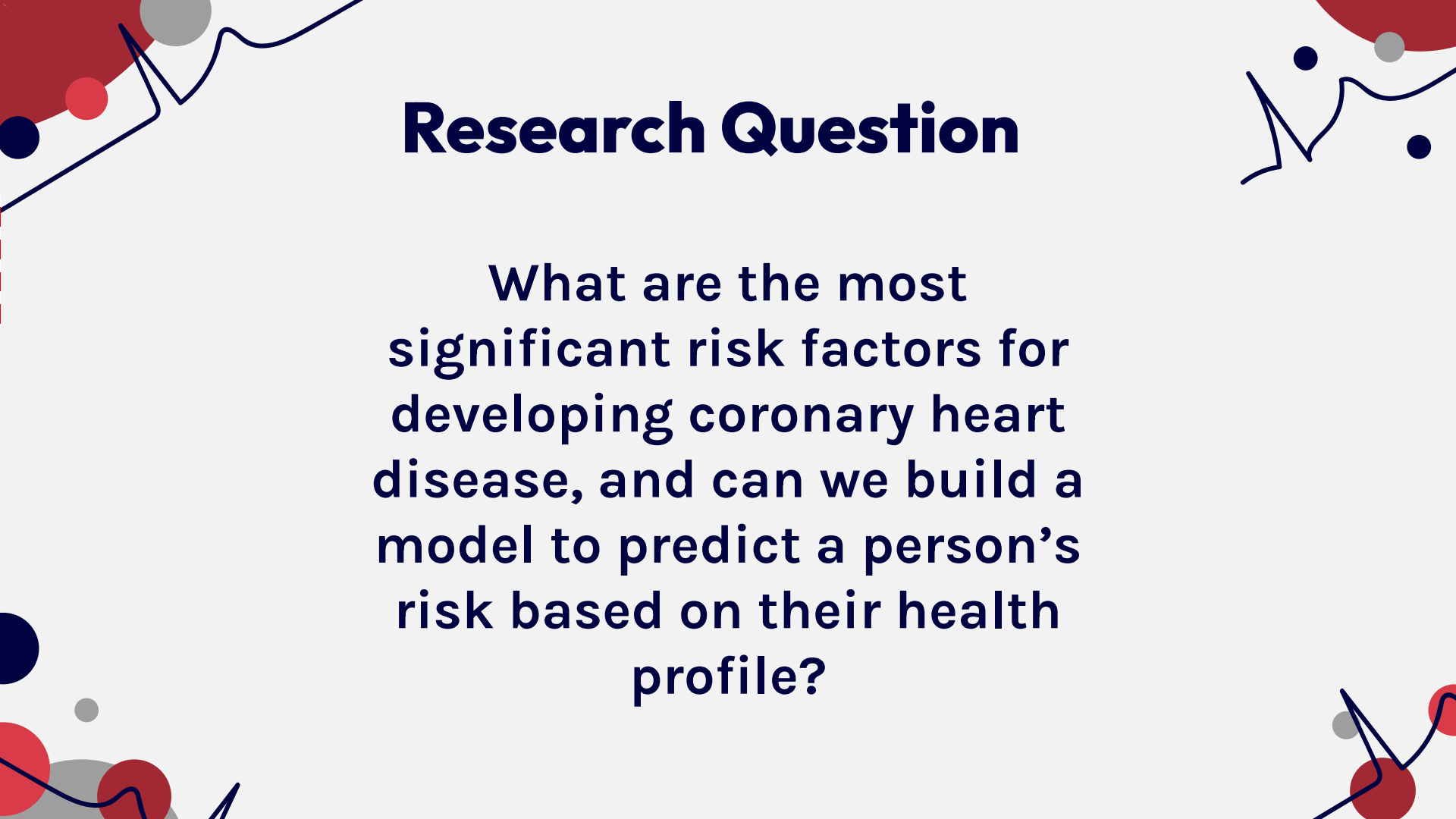
# Predicting Cardiovascular Risk

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# Introduction

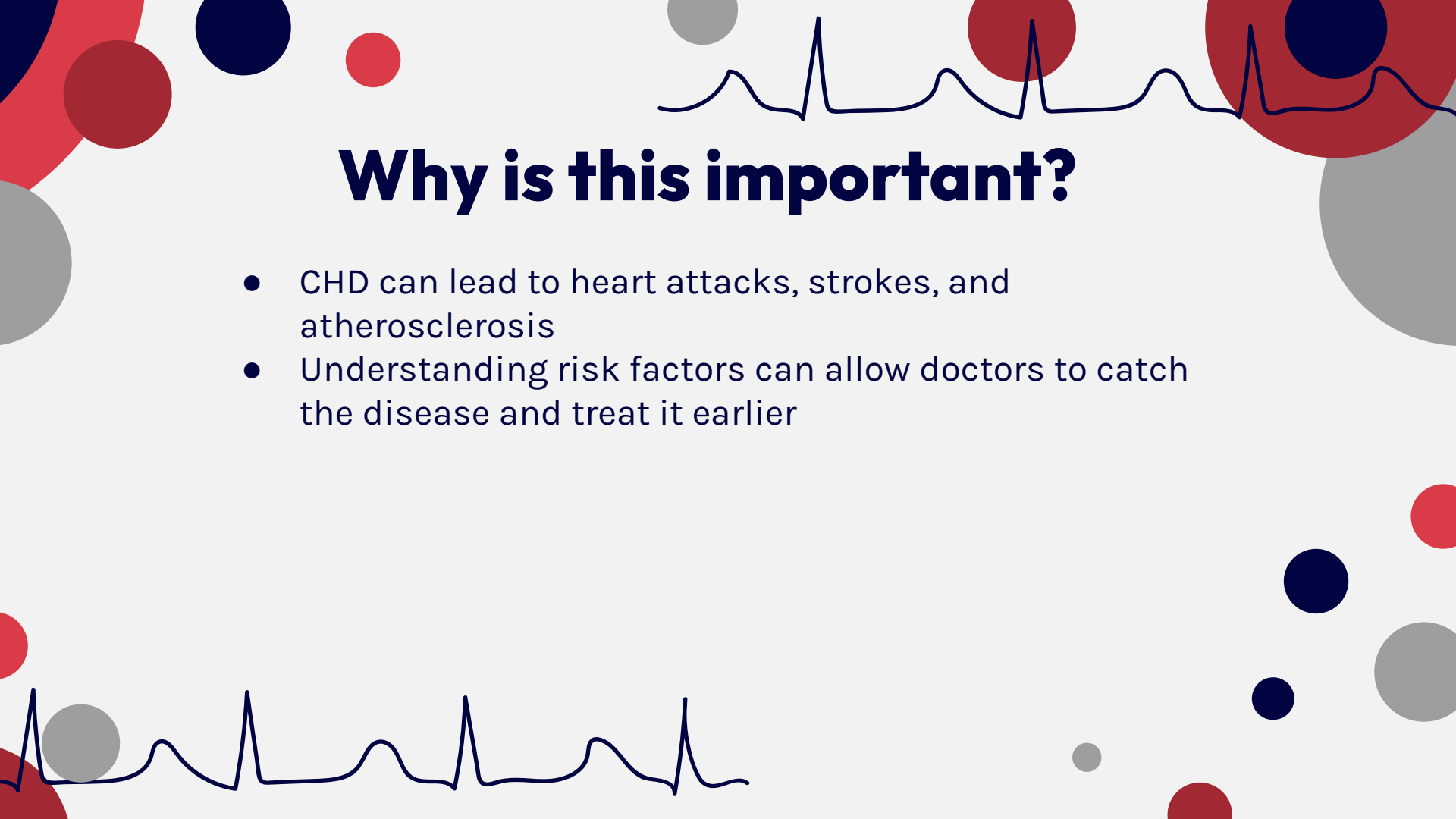
- Framingham Heart Study
- Within this study, there was a focus on Coronary Heart Disease
- In the study, analysis of many factors and their relation to heart disease was covered





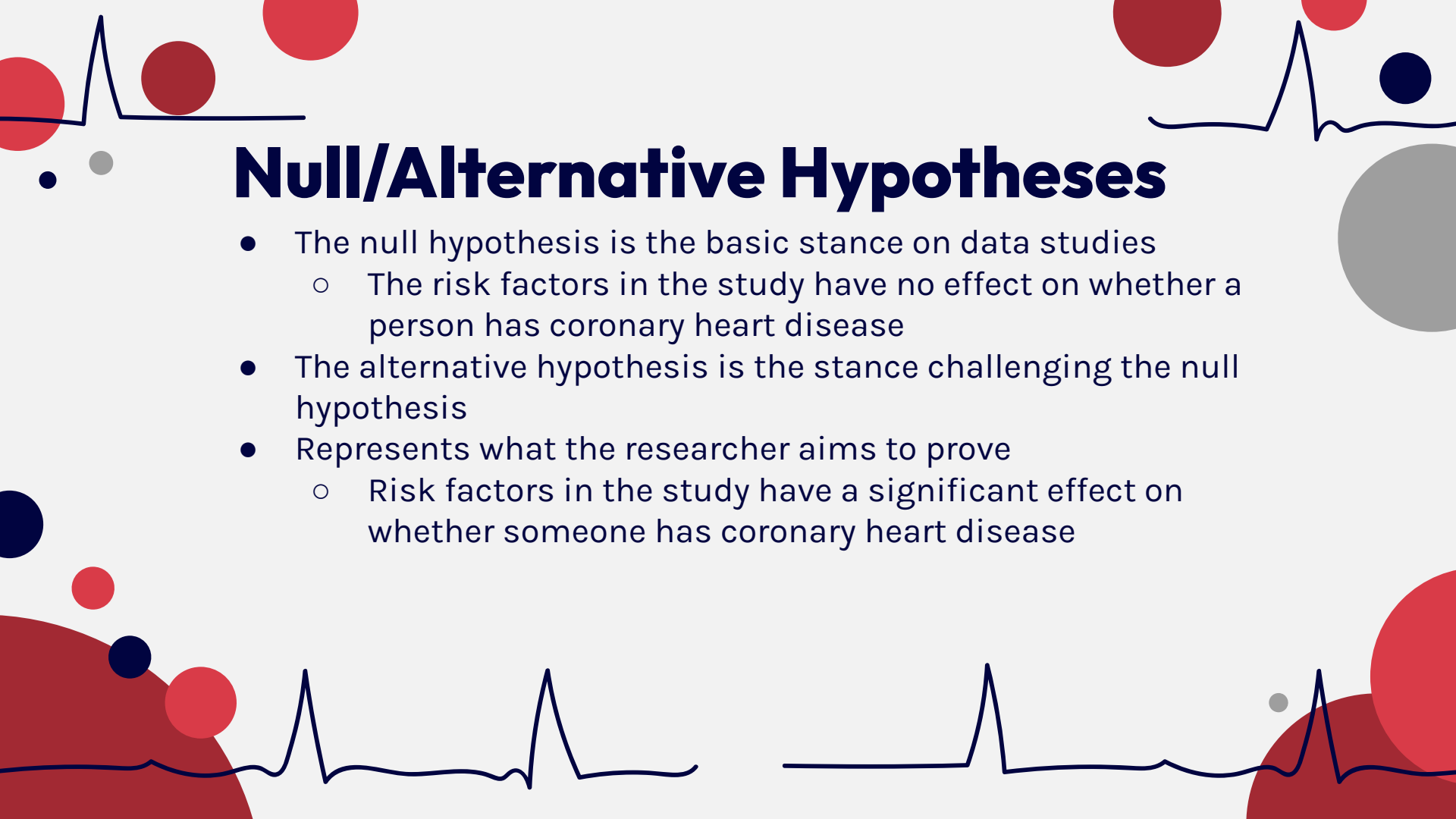
# Research Question

What are the most significant risk factors for developing coronary heart disease, and can we build a model to predict a person's risk based on their health profile?



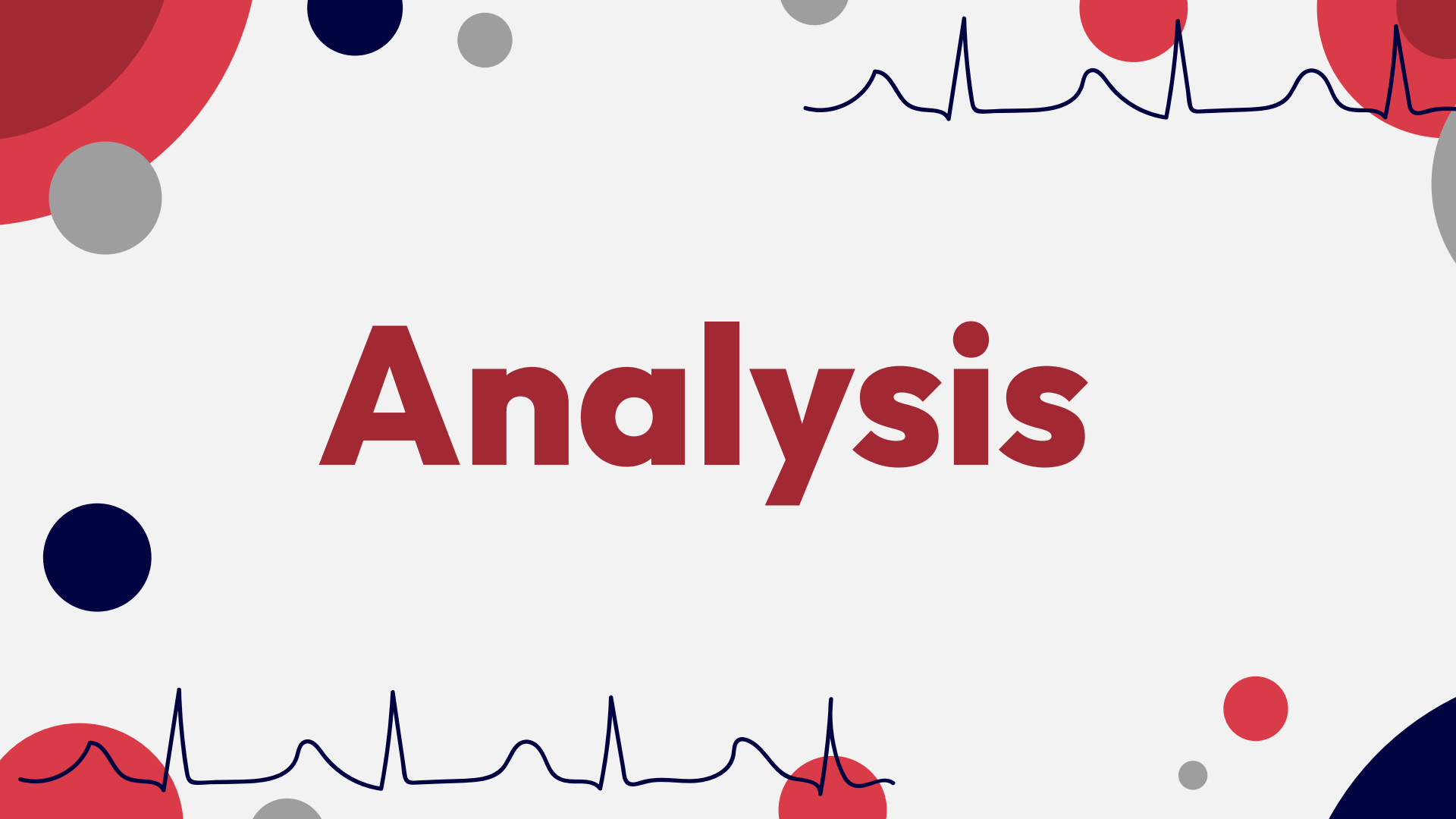
# Why is this important?

- CHD can lead to heart attacks, strokes, and atherosclerosis
- Understanding risk factors can allow doctors to catch the disease and treat it earlier



# Null/Alternative Hypotheses

- The null hypothesis is the basic stance on data studies
  - The risk factors in the study have no effect on whether a person has coronary heart disease
- The alternative hypothesis is the stance challenging the null hypothesis
- Represents what the researcher aims to prove
  - Risk factors in the study have a significant effect on whether someone has coronary heart disease

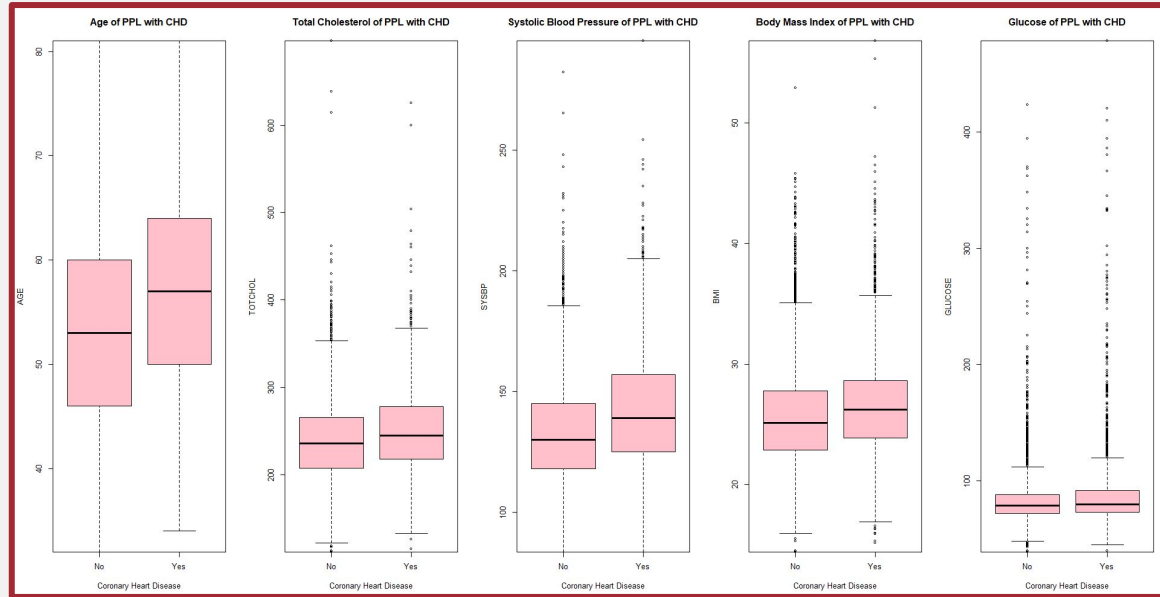
The background is a light gray color. It is decorated with several large, semi-transparent circles in red and dark blue. A dark blue line graph, resembling a heartbeat or a signal waveform, is positioned horizontally across the top and bottom of the image. The word "Analysis" is centered in the middle of the image in a bold, red, sans-serif font.

# Analysis

# Key Variables

| Variables                                                             | Type of Variable | Role in Analysis |
|-----------------------------------------------------------------------|------------------|------------------|
| ANYCHD: Did the patient develop coronary heart disease? (0=No, 1=Yes) | Categorical      | Target           |
| AGE: Age at the time of examination.                                  | Numerical        | Predictor        |
| SEX: Participant's sex (1=male; 2=female).                            | Categorical      | Predictor        |
| TOTCHOL: Total cholesterol (mg/dL).                                   | Numerical        | Predictor        |
| SYSBP: Systolic blood pressure (mm Hg).                               | Numerical        | Predictor        |
| BMI: Body Mass Index                                                  | Numerical        | Predictor        |
| CURSMOKE: Current smoking status (0=no; 1=yes).                       | Categorical      | Predictor        |
| DIABETES: Diabetic status (0=no; 1=yes).                              | Categorical      | Predictor        |
| GLUCOSE: Glucose level (mg/dL)                                        | Numerical        | Predictor        |

# Box Plots

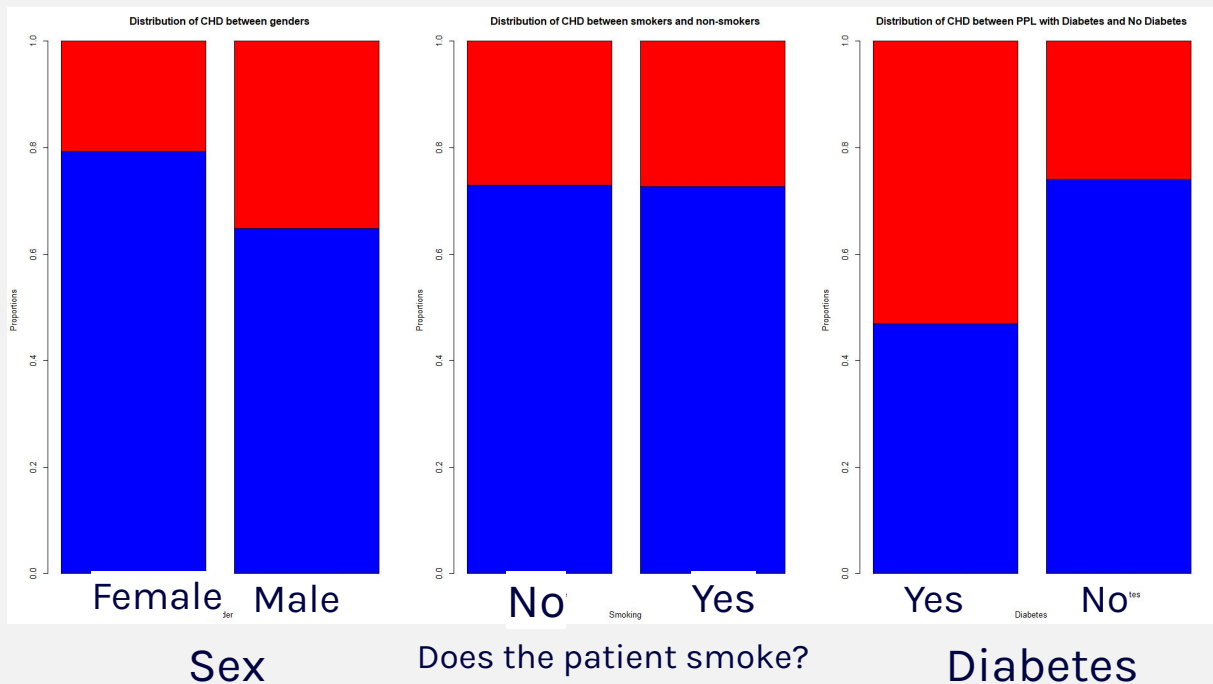


**Age has the highest proportional difference between medians of the numerical variables**



# Proportional Bar Plots

Key:  
Blue = No CHD  
Red = CHD



**Diabetes has the highest proportional difference of the categorical variables.**

# Two Sample T-Test

```
> t.test(TOTCHOL ~ anychd_label, data = heart_data_clean)

Welch Two Sample t-test

data: TOTCHOL by anychd_label
t = -10.732, df = 4591.3, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
95 percent confidence interval:
 -13.273882 -9.173331
sample estimates:
mean in group No mean in group Yes
    238.5252         249.7488
```

- A t-test is a statistical hypothesis test used to compare the means of two groups, or to compare a sample mean to a known population mean.
- The difference of the average total cholesterol between patients with and without coronary heart disease is 11.22 (mg/dl).

# Chi-squared Tests

```
> chisq.test(table(heart_data_clean$diabetes_label, heart_data_clean$anychd_label))
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(heart_data_clean$diabetes_label, heart_data_clean$anychd_label)
```

```
X-squared = 150.38, df = 1, p-value < 2.2e-16
```

```
> chisq.test(table(heart_data_clean$sex_label, heart_data_clean$anychd_label))
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(heart_data_clean$sex_label, heart_data_clean$anychd_label)
```

```
X-squared = 261.27, df = 1, p-value < 2.2e-16
```

- A Chi-squared test is a statistical test used to determine if there's a significant relationship between two categorical variables
- Both diabetes and sex are significant in predicting whether a patient has coronary heart disease according to the Chi-squared test.

# Linear Probability Model

```
> lpm_SYSBP <- lm(ANYCHD ~ SYSBP, data = heart_data_clean)
> summary(lpm_SYSBP)
```

Call:

```
lm(formula = ANYCHD ~ SYSBP, data = heart_data_clean)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -0.8088 | -0.2789 | -0.2163 | 0.5629 | 0.9217 |

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )   |
|-------------|------------|------------|---------|------------|
| (Intercept) | -0.2289268 | 0.0265521  | -8.622  | 2e-16 ***  |
| SYSBP       | 0.0036798  | 0.0001927  | 19.100  | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4368 on 10019 degrees of freedom

Multiple R-squared: 0.03513, Adjusted R-squared: 0.03504

F-statistic: 364.8 on 1 and 10019 DF, p-value: < 2.2e-16

- A type of regression model used when the dependent variable is binary (0 or 1), and the relationship between the independent variables and the probability of the outcome is assumed to be linear
- Systolic Blood Pressure has the biggest impact on the overall model of patients with and without coronary heart disease

# Multiple Linear Probability Model

```
> lpm_data <- lm(ANYCHD ~ AGE + SEX + TOTCHOL + SYSBP + BMI + CURSMOKE + DIABETES + GLUCOSE, data = heart_data_clean)
> summary(lpm_data)
```

Call:  
lm(formula = ANYCHD ~ AGE + SEX + TOTCHOL + SYSBP + BMI + CURSMOKE + DIABETES + GLUCOSE, data = heart\_data\_clean)

Residuals:

|  | Min     | 1Q      | Median | 3Q     | Max    |
|--|---------|---------|--------|--------|--------|
|  | -0.8261 | -0.2984 | 0.1789 | 0.4854 | 1.0639 |

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | -4.816e-01 | 4.731e-02  | -10.180 | < 2e-16 ***  |
| AGE         | 4.711e-03  | 5.017e-04  | 9.390   | < 2e-16 ***  |
| SEX         | -1.540e-01 | 8.764e-03  | -17.574 | < 2e-16 ***  |
| TOTCHOL     | 9.329e-04  | 9.601e-05  | 9.717   | < 2e-16 ***  |
| SYSBP       | 2.379e-03  | 2.119e-04  | 11.227  | < 2e-16 ***  |
| BMI         | 6.108e-03  | 1.110e-03  | 5.504   | 3.80e-08 *** |
| CURSMOKE    | 3.272e-02  | 9.043e-03  | 3.618   | 0.000298 *** |
| DIABETES    | 1.837e-01  | 2.449e-02  | 7.504   | 6.73e-14 *** |
| GLUCOSE     | 8.704e-05  | 1.990e-04  | 0.438   | 0.661756     |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4233 on 10012 degrees of freedom  
Multiple R-squared: 0.09458, Adjusted R-squared: 0.09386  
F-statistic: 130.7 on 8 and 10012 DF, p-value: < 2.2e-16

- A type of regression analysis used when the dependent variable is binary (0 or 1), and you want to see how multiple independent variables affect the probability of the outcome
- According to the multiple linear probability model, all the variables combined result in a total impact of 9.4% of the overall model of patients with and without CHD.
- All the variables break the null hypothesis except glucose

# Conclusion

- The most significant risk factors for developing coronary heart disease are systolic blood pressure, total cholesterol, and diabetes.
- We can build a model to predict a person's risk based on their health profile.
- All the variables except glucose are relevant to the model because their p-values are all really small, meaning that the variable is more likely to reject the null hypothesis theory.



**Thank You For  
Listening**